

Chapter

Geometric Progression

1. The 4th, 6th the last term of a geometric progression are 10, 40 and 640, respectively. If the common ratio is positive, find the first term, common ratio and the number of terms of the series.

Main 2020

2. The sum of the first 3 terms of a GP is 1 and the sum of the next 3 terms is 27. Determine the first term, the common ratio and the sum of the GP to n terms.

Comp 2020

3. The first and last term of a Geometric Progression (GP) are 3 and 96 respectively. If the common ratio is 2, then find

(i) n in the number of terms of the GP.

(ii) Sum of the n terms.

Main 2019

4. If the fourth, seventh and tenth terms of a GP. are x, y, z respectively, prove that x, y, z are in GP

Comp 2019

Solution

1.

Ans: Let the first term and common ratio of GP be a and r , respectively and total number of terms be n .

According to the question, we have

$$\begin{aligned} \text{4th term} \quad a_4 &= ar^{4-1} \\ 10 &= ar^3 \end{aligned} \quad (1)$$

$$\begin{aligned} \text{6th term} \quad a_6 &= ar^{6-1} \\ 40 &= ar^5 \end{aligned} \quad (2)$$

$$\begin{aligned} \text{and last term} \quad a_n &= ar^{n-1} \\ 640 &= ar^{n-1} \end{aligned} \quad (3)$$

Dividing Eq. (2) by Eq. (1), we get

$$4 = r^2$$

Since common ratio is positive,

$$r = 2$$

Now substituting $r = 2$ in eq (1) we get

$$a = \frac{10}{8} = \frac{5}{4}$$

Substituting value of a and r in eq (3), we get

$$\begin{aligned} 640 &= \frac{5}{4} 2^{n-1} \\ 640 \times \frac{4}{5} &= 2^{n-1} \\ 512 &= 2^{n-1} \\ 2^9 &= 2^{n-1} \\ n - 1 &= 9 \end{aligned}$$

$$n = 10$$

First terms is $\frac{5}{4}$, common ration is 2 and number of term of the series is 10.

2.

Ans:

Let a be the first term and r be the common ratio. Then we can write first three term as a , ar and ar^2 and next three term ar^3 , ar^4 and ar^5 .

Since sum of first 3 terms of GP is 1, thus

$$\begin{aligned} a + ar + ar^2 &= 1 \\ a(1 + r + r^2) &= 1 \end{aligned} \quad \dots(1)$$

Since sum of next 3 terms of GP is 27, thus

$$\begin{aligned} ar^3 + ar^4 + ar^5 &= 27 \\ ar^3(1 + r + r^2) &= 27 \end{aligned} \quad \dots(2)$$

Dividing eq (2) by eq (1) we have

$$\begin{aligned} \frac{ar^3(1 + r + r^2)}{a(1 + r + r^2)} &= \frac{27}{1} \\ r^3 &= 27 \\ r &= 3 \end{aligned}$$

Now substituting $r = 3$ in eq (1) we have

$$\begin{aligned} a(1 + 3 + 3^2) &= 1 \\ a(1 + 3 + 9) &= 1 \\ a(13) &= 1 \\ a &= \frac{1}{13} \end{aligned}$$

Now

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{1}{13} \left[\frac{3^n - 1}{3 - 1} \right]$$

$$= \frac{1}{26} (3^n - 1)$$

3.

Ans:

First term of the GP,

$$a = 3$$

Last term of the GP,

$$a_n = 96$$

Common ratio,

$$r = 2$$

(i) Number of terms

$$a_n = ar^{n-1}$$

$$96 = 3(2)^{n-1}$$

$$32 = 2^{n-1}$$

$$2^5 = 2^{n-1}$$

$$n - 1 = 5 \Rightarrow n = 6$$

Number of terms = 6

(ii) Sum of the n terms

$$S_n = \frac{a(r^n - 1)}{r - 1}, \quad r > 1$$

$$S_6 = \frac{3(2^6 - 1)}{2 - 1}$$

$$= 3(64 - 1)$$

$$= 3 \times 63$$
$$= 189$$

Correct answers:

$n = 6, \quad \text{Sum} = 189$

4.

Ans:

Let a be the first term and r be the common ratio of the given GP.

Given:

$$a_4 = x, \quad a_7 = y, \quad a_{10} = z$$

Using the formula $a_n = ar^{n-1}$,

$$a_4 = ar^3 = x$$

$$a_7 = ar^6 = y$$

$$a_{10} = ar^9 = z$$

If x, y, z are in GP, then

$$y^2 = xz$$

Now,

$$y^2 = (ar^6)^2 = a^2r^{12}$$

and

$$xz = (ar^3)(ar^9) = a^2r^{12}$$

$$\text{LHS} = \text{RHS}$$

Hence, x, y and z are in GP.
